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Assessment cover

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Module No:	COMP6011	Module title:	Machine Learning

Assessment number:	**CW1**	Assessment title:	**Performance Comparison of Machine Learning Algorithms**

Banner assignment identifier	*CWS1WEEK05*	Due date and time**:**	**October 24, 2025, 1pm**

Estimated total time to be spent on assignment:	*20 hours per student direct work on coursework, plus 22 hours per student on independent study*

LEARNING OUTCOMES

On successful completion of this assignment, students will be able to achieve the following learning outcomes (LOs): *LO numbers and text to be copied and pasted from the module handbook*
Critically evaluate the requirements for application of system intelligence in an autonomous system.
Build independence, confidence and advanced professional awareness skills within the discipline including ethical, social, legal and risk considerations.
Synthesise and independently develop complex documentation to support autonomous systems created.

Statement of Compliance
By submitting this assessment I declare that the work submitted is my own and that the work I submit is fully in accordance with the University regulations regarding assessments.
(www.brookes.ac.uk/uniregulations/current)

Use of AI Tools: You are required to use this form to declare which AI tools you have used and how you have used them. Please complete the form and attach it to your submission as an Appendix, if you have used such tools.

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❖ Part A

- **A**

Data with three or more probable categories falls under multi-class classification. K-Nearest Neighbours **(KNN)** and Naïve Bayes are the two algorithms that support this, but support comes in different ways. The basic principle of **KNN** is to assign an observation to a class which is most frequent among its nearest neighbours by considering distance as a measure of similarity. This makes the method very flexible, but at times it may turn out to be highly sensitive in cases of overlapping or imbalanced data.
Naïve Bayes calculates the probability for each class then selects the highest one. The independence assumption makes calculations easy and works well even with many classes. Briefly, **KNN** depends on similarity. Naïve Bayes depends on probability-going two different ways to reach the same end.

- **B - I**

The Heart Failure Clinical Records **(HFCR)** data set is composed of 299 patients described by twelve attributes and a binary outcome variable *'death_event'* (0 = survived, 1 = deceased). All the variables are numeric with no missing values, thus clean and consistent to work with. There are 13 columns in this data set which makes it a good choice for supervised learning problems, especially for algorithms like **KNN** and **Naïve Bayes**.

- **B - II**

To check data quality, the dataset was reviewed for missing or null values.
No gaps were found in any of the 13 columns, confirming that the data are complete and ready for analysis without any cleaning or imputation.

- **B – III**

Descriptive statistics of all variables were analyzed to know their distribution and variation. The mean age of patients is 60.8 years (between 40-95 years), and the mean ejection fraction is 38%. There is wide variation in these and certain other features, such as serum creatinine and creatine phosphokinase which have extremely high maximum values-who are possible outliers. Binary variables-these take on values of 0 or 1 to indicate the absence or presence of conditions such as anaemia, diabetes, and high blood pressure.

The dataset, all things considered, shows a pretty realistic and varied clinical population.

The histograms show that continuous variables such as age and ejection fraction follow near-normal distributions, while serum creatinine and creatinine phosphokinase are strongly right-skewed with a few extreme cases. Boxplots confirmed these outliers, especially in creatinine phosphokinase, platelets, and serum creatinine, which likely represent severe clinical cases

rather than errors, so they were kept in the dataset.

Finally, the target variable *'death_event'* shows about 68% survivors and 32% deceased patients, a moderate but acceptable class imbalance typical in medical data.
This allows standard accuracy metrics to be used reliably in later analysis.

- **B - IV**

Before model training, the dataset was divided into **i**ndependent features **(X)** and the target variable **(Y)**.
The matrix **X** contains 12 clinical characteristics for each of the 299 patients, while **Y** represents the binary outcome 'death_event'.

To ensure all features contribute equally to the model, **Z - score normalisation** was applied using Standard scaler from *scikit-learn*. This transformation centres each feature by subtracting its mean and scaling it to unit variance:

$$z = \frac{x - \mu}{\sigma}$$

The process guarantees that no feature dominates others due to differences in scale.
For instance, features such as platelets (range ≈ 25000–850000) would otherwise overshadow smaller-scale features like serum_creatinine (range ≈ 0.5–9.4).

After scaling, all numerical variables had a mean close to 0 and a standard deviation close to 1, confirming successful normalisation.

This step is essential, particularly for distance-based models such as **KNN.**

- **Part B**

To facilitate model training, validation and test, the dataset was divided into 3 subsets. Applying from scikit-learn, train_test_split carried out the split by using 70% of the data for training and the remaining 30% equally for validation and test (15% each). The split also stratified by the target variable 'deatch_event' to reflect the same class proportions (approximately 68% survivors and 32% deceased). To ensure data leakage was mitigated, the scaler was only fitted on the training data, subsequently applied to the other subsets while keeping the parameters the same.

During **KNN** tuning, different **k** values (1, 3, 5, 7, 9) were considered. The best validation accuracy was achieved at k = 5 (75.5%) and test accuracies for k = 3, 5, and 7 yielded 68.9%, 64.4% and, 73.3% respectively. Thus, the difference in validation accuracy and test accuracy is trivial and expected, as **KNN** performs sensitivity to local patterns in the data. In summary, the best generalisation between bias and variance is between k=5-7, and therefore k = 5 was selected as the best performance for the final model.

- **Part C**

A 5-fold cross-validation was executed with the training data to test the stability of the **KNN** model and validate its best value of **k**. For each fold, the model was trained

with 80 % of the data and validated with the other 20 %. Mean training and validation errors for each k = 1, 3, 5, 7, 9 were recorded. The results indicate that the minimum validation error (0.2436) happebs at k = 5, which is the same value that was generated from the single validation in Part B. As K increases, the training error increases and the validation error decreases and then stabilises, as is explained in Lecture 2 **(KNN)** with the classical bias–variance trade-off. This pattern of error suggests that k = 5 generates the best generalisation of the model, as it best balances underfitting and overfitting. Therefore, it was chosen as the k* for comparision with Naïve Bayes.

- **Part D**

The Gaussian **Naïve Bayes** classifier was trained on the normalised features to predict patient survival. This algorithm assumes conditional independence between features and uses a Gaussian distribution for each attribute, which is appropriate given the continuous and approximately normalised data. Both **KNN** (k = 7) and **Naïve Bayes** had the same validation accuracy (75.6 %), but **Naïve Bayes** performed better on the test set (71.1 % vs. 64.4 %), suggesting it generalises better. **KNN** had a higher training accuracy (81.3 %), indicating some mild overfit, but **Naïve Bayes** produced more consistent results through the datasets. The differences can be explained by the nature of the model:

- The **KNN** algorithm relies on local distances and therefore is sensitive to data density and scaling

- **Naïve Bayes** models global distributions of features and performs well on small datasets with independent features, the correlated features of the HFCR dataset is an indication of Naïve Bayes' capabilities.

Thus, we conclude that Naïve Bayes is simpler, faster and robust for this dataset even though KNN is a little more flexible but also has some potential to overfit. As stated in Lecture 3 - Naïve Bayes Classifier, this algorithm frequently performs equally well or better than more complex algorithms in situations where the independence assumption approximately holds. The HFCR data has low correlation for almost all features (refer to the correlation matrix) which further indicates the potential of an independent feature assumption with the Naïve Bayes approach.

- **Part E**

Principal Component Analysis (PCA) was applied to the Heart Failure Clinical Records dataset to reduce dimensionality and remove redundant features.
The **scree plot** of cumulative explained variance showed that **11 principal components** were sufficient to retain **95% of the total variance**.
This means that almost all the useful information in the original 12 features is preserved while simplifying the dataset, which helps to reduce noise and improve model efficiency.

After transforming the data using these 11 components, both

classifiers from earlier parts were re-evaluated.
For **KNN**, performance improved slightly, with validation accuracy increasing from 0.755 to 0.777, and test accuracy from 0.644 to 0.688.
For **Naïve Bayes**, validation accuracy also rose from 0.755 to 0.777, and test accuracy from 0.711 to 0.733.
These results indicate that reducing the number of features helped both models, particularly **KNN**, which is sensitive to high-dimensional spaces.
Naïve Bayes also benefited slightly, as **PCA** reduced feature correlations, making its independence assumption more valid.
In conclusión, **PCA** improved generalisation and simplified both models without losing relevante information.

❖ Conclusion

This project let us to explored how machine learning techniques can predict heart failure outcomes using real clinical data.
Through the use of KNN, Naïve Bayes, and PCA, it became clear that even simple models can deliver solid results when the data are well-prepared and properly validated.
Among all tested approaches, Naïve Bayes with PCA achieved the best balance between accuracy and generalisation.

❖ References

- Scikit-learn Documentation – KNN Classifier: https://scikit-learn.org/stable/modules/neighbors.html
- Scikit-learn Documentation – Naive Bayes Models: https://scikit-learn.org/stable/modules/naive_bayes.html
- Scikit-learn Documentation – Decomposition: PCA: https://scikit-learn.org/stable/modules/decomposition.html#pca
- UCI Machine Learning Repository – Heart Failure Clinical Records: https://archive.ics.uci.edu/dataset/519/heart+failure+clinical+records
- Chicco, D., & Jurman, G. (2020). *Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone.:* https://doi.org/10.1186/s12911-020-1023-5

❖ Tables/Figures

- Part A: 299 Samples and 13 columns.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 299 entries, 0 to 298
Data columns (total 13 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   age                       299 non-null    float64
 1   anaemia                   299 non-null    int64
 2   creatinine_phosphokinase  299 non-null    int64
 3   diabetes                  299 non-null    int64
 4   ejection_fraction         299 non-null    int64
 5   high_blood_pressure       299 non-null    int64
 6   platelets                 299 non-null    float64
 7   serum_creatinine          299 non-null    float64
 8   serum_sodium              299 non-null    int64
 9   sex                       299 non-null    int64
 10  smoking                   299 non-null    int64
 11  time                      299 non-null    int64
 12  death_event               299 non-null    int64
dtypes: float64(3), int64(10)
memory usage: 30.5 KB
```

	age	anaemia	creatinine_phosphokinase	diabetes	ejection_fraction	high_blood_pressure	platelets	serum_creatinine	serum_sodium	sex	smoking	time	death_event
0	75.0	0	582	0	20	1	265000.00	1.9	130	1	0	4	1
1	55.0	0	7861	0	38	0	263358.03	1.1	136	1	0	6	1
2	65.0	0	146	0	20	0	162000.00	1.3	129	1	1	7	1
3	50.0	1	111	0	20	0	210000.00	1.9	137	1	0	7	1
4	65.0	1	160	1	20	0	327000.00	2.7	116	0	0	8	1

- Part A: There are not missing values.

	0
age	0
anaemia	0
creatinine_phosphokinase	0
diabetes	0
ejection_fraction	0
high_blood_pressure	0
platelets	0
serum_creatinine	0
serum_sodium	0
sex	0
smoking	0
time	0
death_event	0

- Part A: Basic Statistics.

	mean	median	std	min	max
age	60.833893	60.0	11.894809	40.0	95.0
anaemia	0.431438	0.0	0.496107	0.0	1.0
creatinine_phosphokinase	581.839465	250.0	970.287881	23.0	7861.0
diabetes	0.418060	0.0	0.494067	0.0	1.0
ejection_fraction	38.083612	38.0	11.834841	14.0	80.0
high_blood_pressure	0.351171	0.0	0.478136	0.0	1.0
platelets	263358.029264	262000.0	97804.236869	25100.0	850000.0
serum_creatinine	1.393880	1.1	1.034510	0.5	9.4
serum_sodium	136.625418	137.0	4.412477	113.0	148.0
sex	0.648829	1.0	0.478136	0.0	1.0
smoking	0.321070	0.0	0.467670	0.0	1.0
time	130.260870	115.0	77.614208	4.0	285.0
death_event	0.321070	0.0	0.467670	0.0	1.0

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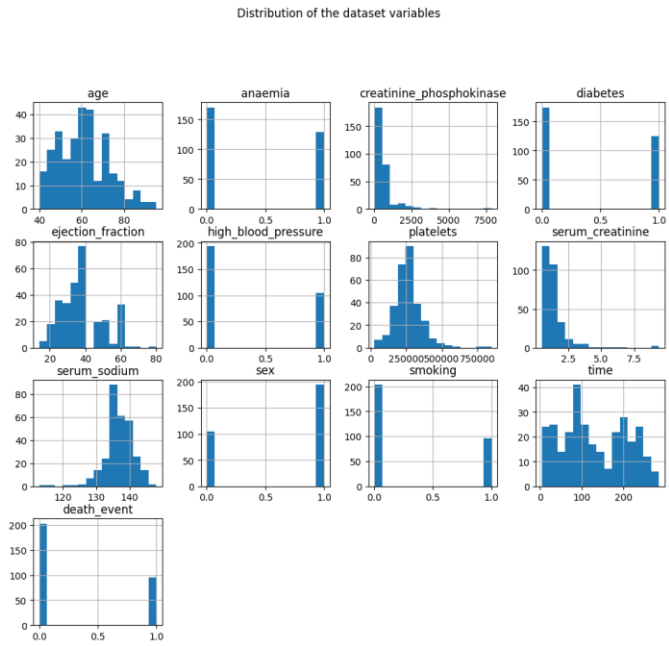
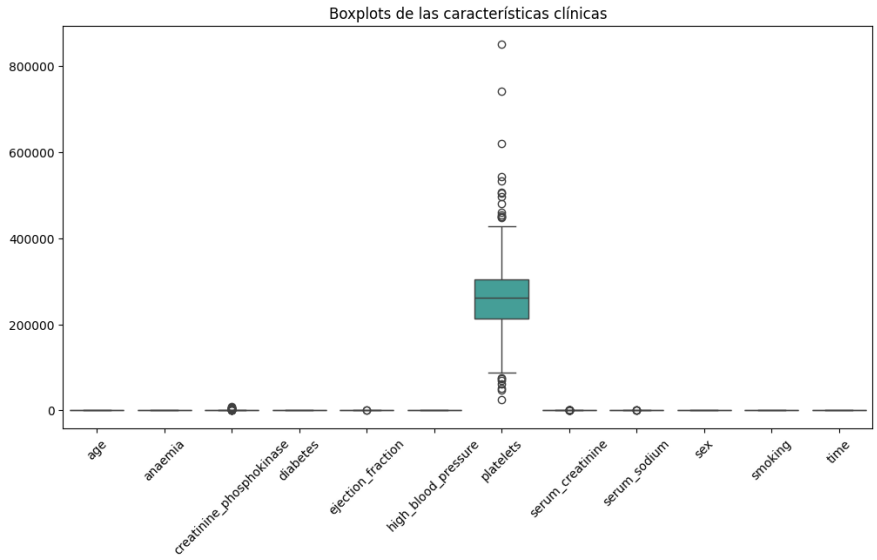
- Part A: Distribution of the Dataset Variables



- Part A: Outliers detector:



- Part A: Percentage of Survival.

death_event	proportion
0	67.892977
1	32.107023

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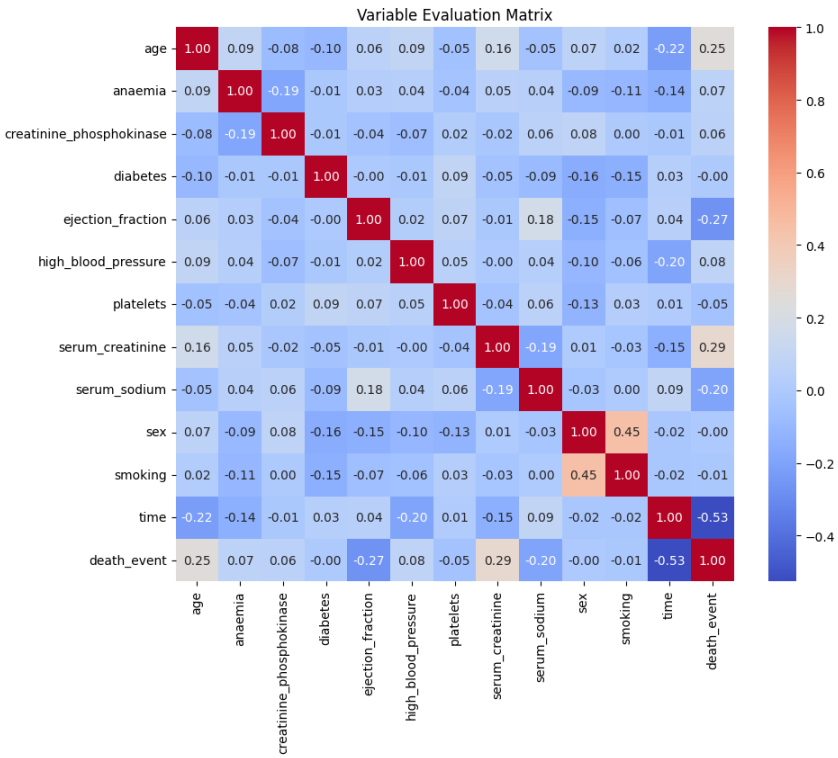
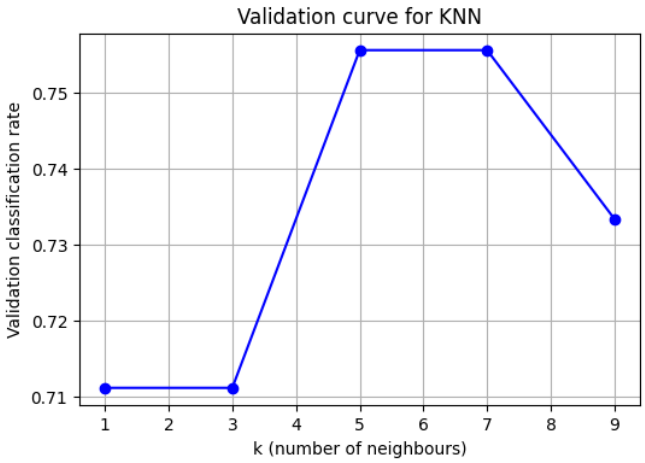
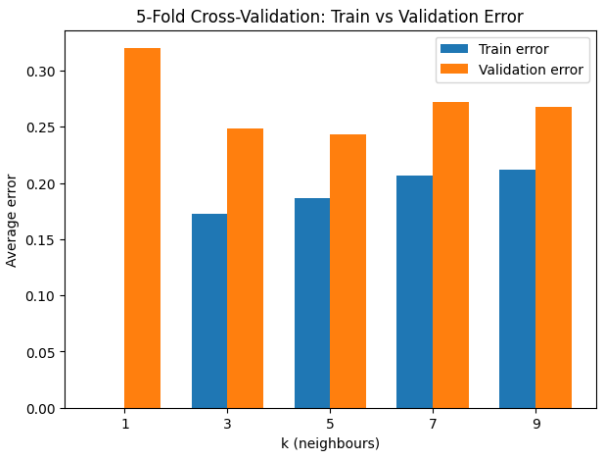
- Part A: Principal Cause of Death.



- Part B: Plot of Validation Curve for KNN.



- Part C: Cross-Validation.



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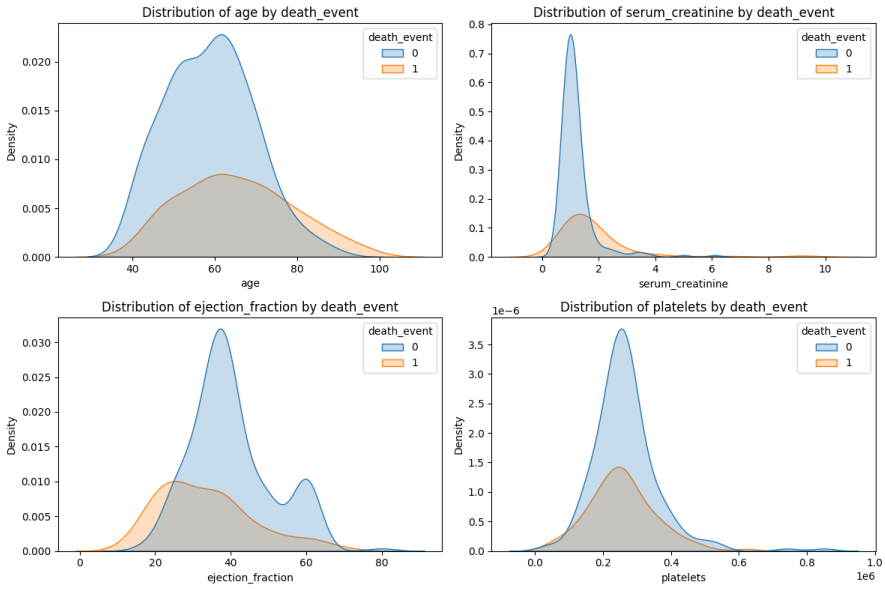
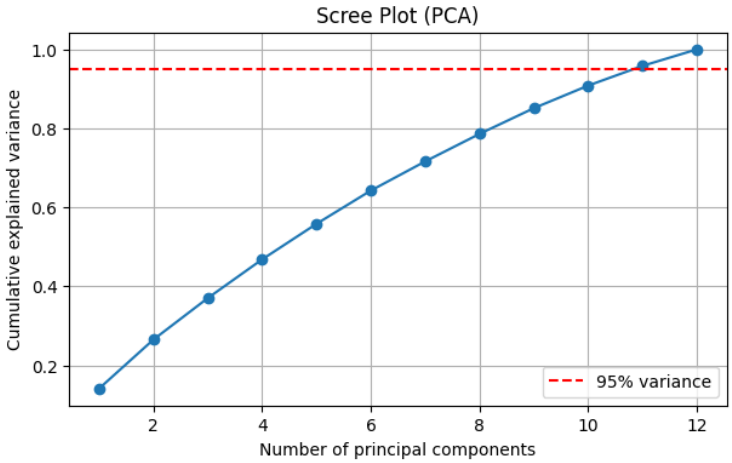
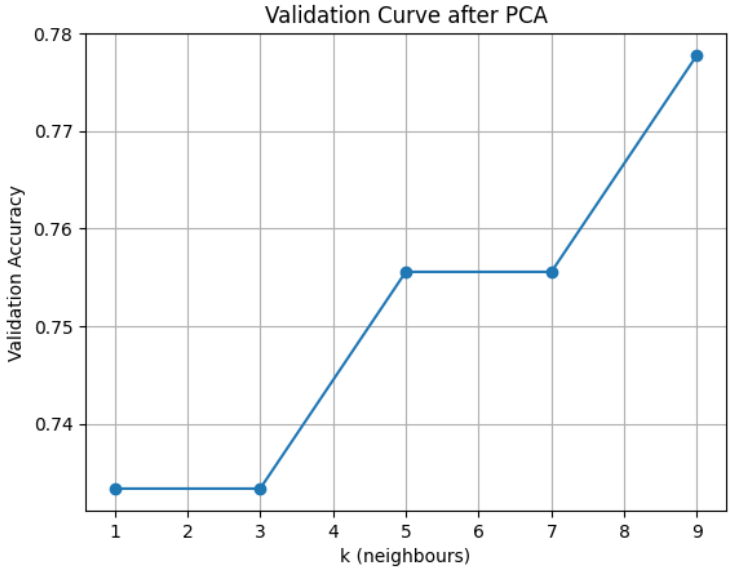
- Part D: Distribution for each class (Naïve Bayes)



- Parte E: Adjust the PCA.



- Part E: Validation Curve after PCA